FINITE NONZERO LENGTH AT A POINT CHARACTERISES THE GENERIC POINTS OF THE SUPPORT OF A COHERENT SHEAF

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: an etale chart translation of support and length into a statement about the stalk, settled by an algebra criterion, followed by a specialisation and soberness argument for the decent case

Editorial note. The label lemma-length-finite names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The length of a coherent module at a point of a locally Noetherian algebraic space and the support of a coherent module, together with the compatibility of both notions with etale base change to a scheme
- The algebra criterion identifying the generic points of the support of a finite module over a Noetherian local ring with the points where the support is the maximal ideal, and finiteness of the module over such a ring
- For the decent case: the lifting of specialisations along an etale morphism between spaces and the fact that the underlying spaces of schemes and of decent spaces are sober

Source reference. Original lemma 00L5: sources/stacks-chow/tags.txt, source lines 775–775. Prerequisite pointer invoked at proof line(s) 599; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 01XZ: sources/stacks-chow/tags.txt, source lines 2538–2538. Prerequisite pointer invoked at proof line(s) 589; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03K5: sources/stacks-chow/tags.txt, source lines 4665–4665. Prerequisite pointer invoked at proof line(s) 611; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03IL: sources/stacks-chow/tags.txt, source lines 4607–4607. Prerequisite pointer invoked at proof line(s) 610; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0BB6: sources/stacks-chow/tags.txt, source lines 14234–14234. Prerequisite pointer invoked at proof line(s) 622; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition 03K6: sources/stacks-chow/tags.txt, source lines 4666–4666. Prerequisite pointer invoked at proof line(s) 613; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 056J: sources/stacks-chow/tags.txt, source lines 6712–6712. Prerequisite pointer invoked at proof line(s) 591; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 01IS: sources/stacks-chow/tags.txt, source lines 1989–1989. Prerequisite pointer invoked at proof line(s) 615; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 07UB: sources/stacks-chow/tags.txt, source lines 9966–9966. Prerequisite pointer invoked at proof line(s) 587; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 07TZ: sources/stacks-chow/tags.txt, source lines 9954–9954. Prerequisite pointer invoked at proof line(s) 605; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme and let X be a locally Noetherian algebraic space over S. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module. Let $x \in |X|$. The following are equivalent*

1. *for some étale morphism $U \to X$ with U a scheme and $u \in U$ mapping to x we have u is a generic point of an irreducible component of $\mathrm{Supp}(\mathcal{F}|_U)$,*
2. *for any étale morphism $U \to X$ with U a scheme and $u \in U$ mapping to x we have u is a generic point of an irreducible component of $\mathrm{Supp}(\mathcal{F}|_U)$,*
3. *the length of $\mathcal{F}$ at x is finite and nonzero.*

If X is decent (equivalently quasi-separated) then these are also equivalent to

4. *x is a generic point of an irreducible component of $\mathrm{Supp}(\mathcal{F})$.*

Proof. Assume $f : U \to X$ is an étale morphism with U a scheme and $u \in U$ maps to x. Then $\mathcal{F}|_U = f^*\mathcal{F}$ is a coherent $\mathcal{O}_U$-module on the locally Noetherian scheme U and in particular $(\mathcal{F}|_U)_u$ is a finite $\mathcal{O}_{U,u}$-module, see Cohomology of Spaces, Lemma 07UB and Cohomology of Schemes, Lemma 01XZ. Recall that the support of $\mathcal{F}|_U$ is a closed subset of U (Morphisms, Lemma 056J) and that the support of $(\mathcal{F}|_U)_u$ is the pullback of the support of $\mathcal{F}|_U$ by the morphism $\mathrm{Spec}(\mathcal{O}_{U,u}) \to U$. Thus u is a generic point of an irreducible component of $\mathrm{Supp}(\mathcal{F}|_U)$ if and only if the support of $(\mathcal{F}|_U)_u$ is equal to the maximal ideal of $\mathcal{O}_{U,u}$. Now the equivalence of (1), (2), (3) follows from by Algebra, Lemma 00L5.

If X is decent we choose an étale morphism $f : U \to X$ and a point $u \in U$ mapping to x. The support of $\mathcal{F}$ pulls back to the support of $\mathcal{F}|_U$, see Morphisms of Spaces, Lemma 07TZ. Also, specializations $x' \rightsquigarrow x$ in $|X|$ lift to specializations $u' \rightsquigarrow u$ in U and any nontrivial specialization $u' \rightsquigarrow u$ in U maps to a nontrivial specialization

$f(u') \rightsquigarrow f(u)$ in $|X|$, see Decent Spaces, Lemmas 03IL and 03K5. Using that $|X|$ and U are sober topological spaces (Decent Spaces, Proposition 03K6 and Schemes, Lemma 01IS) we conclude x is a generic point of the support of $\mathcal{F}$ if and only if u is a generic point of the support of $\mathcal{F}|_U$. We conclude (4) is equivalent to (1).

The parenthetical statement follows from Decent Spaces, Lemma 0BB6. □